# Raj Reddy

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## Education

**University of Utah** **Salt Lake City, UT**
*B.S. Computer Science* *August 2020 – May 2025*

**Relevant Coursework:** Computer Systems, Machine Learning, Computer Graphics, Algorithms, Software Practice I & II, Database Systems, Computer Networks, Foundations of Data Analysis, Models of Computation, and Linear Algebra.

## Skills

**Languages:** Python, TypeScript, C++, SQL, Java

**Software:** Pandas, NumPy, React, Node.js, AWS, Azure, Docker, Next.js, .NET, REST, Django, MongoDB, Linux

## Experience

**L3Harris Technologies** **Dallas, TX**
*Associate Software Engineer* *June 2025 – Present*

- Built containerized microservices for high-frequency sensor data ingestion and storage using Podman and serverless infrastructure, engineering for strict latency and throughput constraints in production environments.
- Developed signal processing algorithms in Python and C++ on embedded Linux systems, applying low-level optimization and profiling to maximize performance under resource constraints.
- Automated CI/CD pipelines with static code analysis and integration testing, maintaining deployment reliability across multiple production services.

**Doxy.me** **Charleston, SC**
*Software Engineer* *August 2024 – May 2025*

- Deployed and operated distributed ML inference pipelines on AWS SageMaker and Fargate, owning reliability, observability, and zero-downtime releases across multiple production services.
- Built TypeScript APIs and WebRTC integrations powering real-time clinical video workflows, designing data models and API contracts consumed by thousands of active providers.
- Owned YAML-based CI/CD configuration and cloud infrastructure deployments, enabling reproducible environments and fast, confident releases across the stack.

**University of Utah** **Salt Lake City, UT**
*Undergraduate Research Assistant – FuTURES Lab* *May 2024 – Jun 2025*

- Analyzed and optimized large-scale datasets, enhancing software configuration testing with gcov and CMake, increasing code coverage by 30% on real-world APIs (Libpng, PyTorch).
- Expanded OSS-Fuzz testing coverage using compile-time options to improve the robustness of full-stack libraries.

## Projects

**University Course Planning/Review Platform** *(Class Best):* Built a multi-service data platform enabling student schedule matching, course overlap detection, and behavioral insights. Developed ETL pipelines in Python to process thousands of course records, user preferences, and time-series activity logs. Built a responsive frontend in React + Tailwind with interactive dashboards, heatmaps, and calendar visualizations.
**Configuration Fuzzer:** Developed a configuration fuzzing tool for OSS-Fuzz, automating build generation to identify critical compile-time configurations and improve code coverage, uncovering previously untested code paths and increasing bug detection effectiveness.
**Telehealth Mental Health Diagnosis Tool (Doxy.me):** Built an AI system that analyzes visual, speech, and tonal cues from telehealth sessions using PyTorch and Llama, delivering structured mental health insights to treating physicians via SageMaker and Fargate inference pipelines. Improved diagnostic accuracy using reinforcement learning and WebRTC video integration.